Using Machine Learning Algorithms With *In Situ* Hyperspectral Reflectance Data to Assess Comprehensive Water Quality of Urban Rivers

Jiannan Cai, Jun Chen, *Member, IEEE*, Xianhui Dou, and Qianguo Xing

***Abstract*— Remotely sensed hyperspectral data can support more effective water quality monitoring. Nevertheless, the variability and complexity of urban river water make it hard to retrieve comprehensive water quality characteristics directly, so that most current water quality assessments rely on semiempirical, semianalytical, or bio-optical approaches. In this study, we carried out simultaneous *in situ* hyperspectral data and water quality measurements. We used the 382 hyperspectral data from urban rivers of Zhongshan City in the Pearl River Delta to test how well the random forest (RF) and one-dimensional convolutional neural networks (1D-CNNs) algorithms retrieved the newly established water quality index (WQI). The RF algorithm also identified essential wavelengths for retrieving the WQI. Our results demonstrate that the RF and 1D-CNN algorithms performed well in WQI estimations. The 1D-CNN model performed significantly better than the RF model, especially on high WQI samples. Both models were insensitive to smoothing of the hyperspectral data, showing that the noise of the original hyperspectral reflectance data has a limited impact on the algorithms. In addition, when we used the essential wavelength data (mainly located between 580–590 nm and near 722, 751, 821, and 830 nm) as input data, we achieved better retrieval results. The 1D-CNN model performed the best with an R^2 of 0.87, RMSE of 0.574, and RPIQ of 3.082 when we used the top tenth percentile of the essential wavelength data. This study demonstrates the potential of the 1D-CNN algorithm for hyperspectral data analysis to retrieve comprehensive water quality.**

***Index Terms*— *In situ* hyperspectral, one-dimensional convolutional neural network (1D-CNN), random forest (RF), urban river, water quality index (WQI).**

I. Introduction

TRADITIONAL water quality monitoring, based on laboratory chemical analysis, is costly, time-consuming, and spatially and temporally insufficient [1]. Various sensor networks (using chemical or optical methods) composed of automated sampling and monitoring instruments are used to monitor and assess water quality to mitigate the spatial and temporal shortcomings. With the recent development of hyperspectral technology, hyperspectral reflectance sensors are being used to assess water quality. Compared with multispectral sensors, hyperspectral sensors provide enough information to reconstruct the reflectance spectra from visible to near-infrared regions of water bodies [2], [3], which provides an opportunity for identifying spectrum characteristics and retrieving different water quality parameters. It is already widely accepted that hyperspectral remote sensing is effective at directly retrieving optically active parameters, such as chlorophyll-a, colored dissolved organic matter (CDOM), and total suspended solid (SS) [4]–[6].

Rather than retrieving individual water quality parameter concentration, there is substantial demand for spatial continuously delineating accurate surface water quality levels (SWQLs) to decision-makers [7]. Currently, the comprehensive assessment of SWQLs using remote sensing remains a challenging task due to the optical complexity of urban waters [8]–[10]. Therefore, retrieving water quality comprehensively using remote sensing techniques has scientific and practical value.

The water quality index (WQI) is one of the most widely used methods for surface water among the approaches for evaluating comprehensive water quality [11]. This index is a mathematical approach used to transform large quantities of water characteristics into a single number that summarizes the water quality level [12], [13]. Because the WQI reflects the combined effects of all water quality parameters, it provides a global vision of the overall water quality. Nevertheless, to date, studies on using remote sensing technology to retrieve WQI have been limited [14]. Furthermore, it is accepted that directly estimating WQI from multispectral characteristics is difficult due to the complexity and diversity of spectrum characteristics [15]. Therefore, methods that take advantage of *in situ* hyperspectral technology could be an appropriate direction for WQI quantitative monitoring.

Hyperspectral data with high dimensions provide valuable water quality information, but these data also come with problems such as data redundancy and high processing costs [16]–[18]. Ordinarily, owing to high acquisition costs, the number of environmental monitoring samples (n) is much smaller than the number of hyperspectral bands (p). This

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Jiannan Cai is with the School of Human Settlements and Civil Engineering, Xi'an Jiaotong University, Xi'an 710049, China, and also with the Zhongshan Ecology and Environmental Agency, Zhongshan 528401, China (e-mail: jncai@foxmail.com).

Jun Chen and Xianhui Dou are with the School of Human Settlements and Civil Engineering, Xi'an Jiaotong University, Xi'an 710049, China (e-mail: chenjun@xjtu.edu.cn; 449269165@qq.com).

Qianguo Xing is with the Yantai Institute of Coastal Zone Research, Chinese Academy of Sciences, Yantai 264003, China (e-mail: qgxing@yic.ac.cn).

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typical "small n and large p" data characteristics will lead to the Hughes' phenomenon, which means the retrieval accuracy reduces when the dimensionality of the data increases, while the training samples remain fixed [19]. This phenomenon will greatly affect the model effect, so it is also named "the curse of dimensionality" [20]. Therefore, algorithms which reduce "the curse of dimensionality" without sacrificing important information are critical for successfully using hyperspectral data.

In many studies, researchers have built estimation algorithms using semi-empirical, semi-analytical, and bio-optical models by hyperspectral data for estimating one or limited number of water quality parameter [10]. The semiempirical and semianalytical models establish the relationships between the remote sensing data and water quality parameters by selecting a few characteristic wavebands [21]–[23]. The bio-optical models are based on the theory of water radiative transfer to establish the relationship between the apparent optical properties (AOPs) and the inherent optical properties (IOPs) [24]. Although these models have achieved good results in estimating and monitoring the water quality, the complex bio-optical environment containing different water pollutants makes it challenging to find a specific internal correlation between the spectrum information and the optically inactive water quality parameters. In this case, the traditional linear and nonlinear regression techniques may not extract the deep spectrum features of the hyperspectral data and model such complex nonlinear relationships [10].

To achieve more retrieval accuracy for more comprehensive water quality at the same time dealing with the Hughes' phenomenon, partial least squares regression, support vector machines, and artificial neural networks are popular methods in hyperspectral remote sensing [25], [26]. Over the past decade, two novel machine learning methods have received increased attention: the random forest (RF) algorithm and the convolutional neural networks (CNN) algorithm. The RF and CNN algorithms have several common advantages, including no need to *a priori* determine the initial assumptions, automatically obtaining features instead of using a manual design, straightforward interpretation of outputs, and robustness to overfitting [27]–[29]. In addition, these two algorithms have outstanding merits: the RF assesses variable importance, and the CNN extracts deep features. Numerous studies revealed that the RF and CNN algorithms achieve satisfactory results for classifying and predicting science research [30]–[32]. The RF and CNN algorithms have been successfully applied in remote sensing for water quality parameter detection [33]–[35]. However, applying the RF or CNN algorithm to integrated water quality retrieval using hyperspectral data is still a new field of research.

Our study monitored *in situ* hyperspectral data for urban surface water in Zhongshan City in the Pearl River Delta. We collected above-water spectrum data from July to September 2019 and conducted water sampling and laboratory analysis to obtain the concentrations of several water quality parameters. The main objectives of our study were: 1) to show the *in situ* hyperspectral response to different levels of WQI values of urban rivers in the region; 2) to evaluate the WQI retrieval with hyperspectral data using the RF and 1D-CNN algorithms; and 3) to use variable importance assessment to extract an optimal subset of variables with higher predictive accuracy.

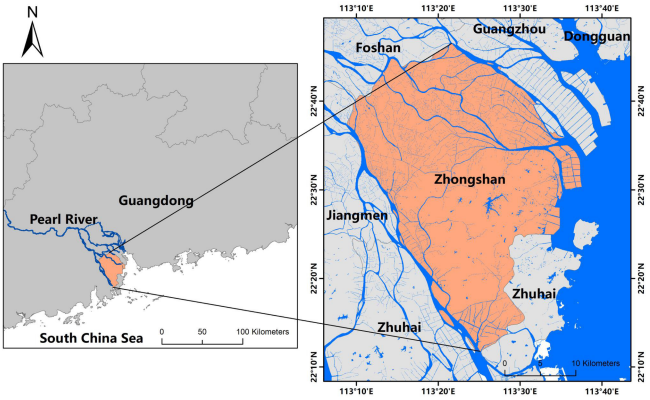


Fig. 1. Location of the study area.

II. Materials and Methods

A. Study Area

Zhongshan City lies in the south-central Pearl River Delta and on the west bank of the Pearl River Estuary. The city has a population of more than 5.6 million. Rivers in Zhongshan are numerous, dense, and perform various functions such as providing a venue for shipping, drinking water supply, a venue for aquaculture, and improving the landscape (Fig. 1). In recent years, the fast-developing economy, intense industrial clusters, and highly developed pond fish farming have created a significant nutrient load to the rivers, leading to a significant decline in water quality.

From 2018 to 2019, the local environment management department comprehensively monitored the water quality, covering more than 310 rivers and gathering over 1200 samplings. The department analyzed up to 19 water parameters for this monitoring project, including chemical oxygen demand (COD), 5-day biochemical oxygen demand (BOD_5), ammonium (NH_4^+-N), total phosphorus (TP), total nitrogen (TN), SS, petroleum, anionic surfactant (LAS), metal and metalloid elements (zinc, copper, iron, cadmium, mercury, lead, arsenic, and selenium), nutrients (NO_2^--N and NO_3^--N), organic pollution parameters (petroleum and volatile phenol), and inorganic pollution parameters (Cl^-, SO_4^{2-}, and F^-). Table I summarizes the water quality observations of the rivers. The table shows that, for the sampling season, the water quality of the different rivers varied greatly. The parameters that exceeded the standards were TP, COD, and NH_4^+-N, which means that the primary source of pollution could have come from adjacent urban regions or livestock waste discharge.

B. Criteria for Comprehensive Water Quality Assessments

Due to the diverse and complex nature of water quality assessment, most criteria for comprehensive assessment use

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TABLE I
SUMMARY OF SIGNIFICANT WATER QUALITY PARAMETERS OF RIVERS IN ZHONGSHAN, 2018–2019

Water quality parameter	Min value	Max value	Mean value	Percentage of samples exceeding standard[a]
NH_4^+-N	N.D.[b]	26.5	4.02±2.19	52.3%
TP	N.D.	8.78	0.39±0.55	41.7%
DO	0.05	10.94	4.02±2.18	33.8%
BOD5	0.5	103	4.2±4.7	16.3%
COD	4.0	195	19.7±17.5	15.5%
LAS	N.D.	7.99	0.12±0.34	4.6%
Petroleum	N.D.	3.9	0.08±0.20	3.2%

[a] Standard reference value: Chinese National Environmental Quality Standard for Surface Water (GB3838-2002), Grade V'.[b] .N.D. means "not detected"



Fig. 2. Photographs of the sampled rivers.

different parameters that depend on the types of water bodies and management objectives. In this study, we aimed to develop a simple but effective way to determine water quality levels. We used the following principles to select critical parameters for the WQI: 1) parameters should represent and reflect the characteristics of river water pollution in an urban area; 2) parameters should have clear standard values for normalization; and 3) the selected parameters should be widely used in routine monitoring on urban rivers. Thus, based on the observations from Table I, we chose NH_4^+-N, TP, and COD to propose the WQI. These parameters are primarily associated with domestic sewage discharge. In addition, we adopted the SS parameter, which is an important indicator of the behavior of pollutants [36], as a measure to calculate WQI. The four parameters we selected are commonly used in urban water quality monitoring and can easily be automatically or manually measured, which benefits effective and low-cost water quality evaluation.

The construction of the WQI requires a normalization step to transform the original value into a dimensionless value. Generally, the reference values for pollution discharge or environmental quality standards for the parameters can be used as the normalization factors [13], [37]. In this study, the normalization factors we used were 30, 1.5, 0.2, and 60 mg/L for COD, NH_4^+-N, TP, and SS, respectively. These values correspond to the Quality Standards for Surface Water Resources (Standard Number: SL-94) Grade 2 value for SS and the Environmental Quality Standards for Surface Water (Standard Number: GB3838-2002) Grade IV values for the other three parameters.

In many procedures, weighting factors that reflect the importance of each parameter are applied to calculate the final WQI. Generally, determining the weights is entirely subjective, leading to significant variation in the evaluation results. Related research shows that the performance of WQI without considering weights in the calculation can also achieve good results [11], [38], [39]. Because we intend to use the standard values as the normalization factors in this study, while the water quality standards are usually formulated in consideration of their impact on water bodies for parameters, a WQI that is determined with clear and universal reference values as normalization factors and computed without weighting factors is a reasonable approach. Therefore, according to the widely used approach of water quality assessment in China's environmental management regulations [40], we defined the WQI equation in this study for each sample was as follows:

$$\mathrm{WQI} = \sum_{i=1}^{n} \frac{C_i}{S_i} \tag{1}$$

where n represents the total number of parameters; C_i represents the measured concentration for each parameter; and S_i represents the reference value.

C. Data Used

1) Water Sampling and Laboratory Analysis: In this study, we simultaneously sampled the water and measured the hyperspectral data from July to September 2019, resulting in 382 samples. The sampling sites were in river and river-like environments in the Zhongshan region and covered a wide variety of optically different surface waters, including main rivers, streams, irrigation ditches, and artificial canals (Fig. 2). The depth of these rivers varies greatly, ranging from about 2.5 m to more than 5 m. The depth of each sampled river far exceeds its transparency, which can be considered as non-optically shallow. In this study, it is considered that the measured spectrum was not affected by the bottom reflectance.

We obtained samples from 30 cm below the water surface and transported the samples to the laboratory to determine the COD, TP, NH_4^+-N, and SS concentration using the standard method suggested by the Chinese National Environmental Quality Standards for Surface Water (Standard Number: GB3838-2002). During analysis, we duplicated all the water sample analyses and applied reagent blanks and certified reference materials for quality assurance and quality control. The accuracy of the certified values and the relative standard deviations of the duplicated analyses were within 10%, indicating excellent reproducibility of the equipment and operations.

2) Reflectance Measurement and Pretreatment: At each site, we measured the *in situ* radiance data at two meters above the water surface within 10 min of the water sampling using an Ocean Optics USB4000 spectrometer (Ocean Optics,

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Inc., Dunedin, FL, USA), which has a spectrum range from 360 to 1000 nm and an interval of ~0.2 nm. The measurements occurred during cloud-free weather and within three hours of solar noon. We calculated water-leaving reflectance R as

$$R = \rho_p \times \frac{L_W}{L_p} \tag{2}$$

where ρ_p is the reflectance of the reference plaque with an approximately constant value of 0.25, L_p represents the radiance received from the reference plaque, and L_w is the water surface radiance.

The radiance sensor pointed to the water surface with an angle of 40° from nadir when measuring. To obtain averaged hyperspectral data, we made three repeated measurements within a short time at each site. During data processing, we calculated only wavelengths from 400 to 900 nm of the spectrum due to the observed low noise-to-signal ratio.

Because the distribution and composition of noise in hyperspectral data are complex, preprocessing techniques are usually applied to achieve noise reduction filtering. Savitzky–Golay filtering is one of the most popular spectrum pretreatments because it simultaneously smooths and differentiates absorption spectra [41]. To verify the impact of data preprocessing (especially the removal of spectrum noise) on the retrieval models, we additionally used the Savitzky–Golay method to preprocess the datasets used as inputs to our models.

D. Machine Learning Algorithm Construction

1) Random Forest Algorithm: The RF algorithm combines tree predictors such that each tree depends on the values of an independently sampled random vector. With the RF algorithm, the training samples were divided into several subsamples (nTrees) using a random sampling method (i.e., a bootstrap method) during the tree building and a random subset of features (mTry) were selected from all the features of each subsample [42]. Each tree independently grows to its maximum size based on a bootstrap sample from the original training dataset without any pruning. Data not in the bootstrap sample (called OOB data) are predicted using the tree grown with the bootstrap sample at each bootstrap iteration. Aggregating the trees' predictions (the average for the regression) occurs during the test sample's prediction process. Since the OOB data are not used in the training process, the prediction error offers an unbiased assessment of the trees' predictive accuracy [43]. The optimal nTrees and mTry values for the RF regression models developed in our study were 1000 and 1/3 of the total number of variables.

The RF algorithm inherently assesses the relative importance of the regression variables, which benefits from evaluating the contribution of the characteristic variables and gaining better insight into the predictions [44]. To determine the importance of a given feature, we computed the OOB error for that variable and permuted the error, while all other variables were left unchanged. Therefore, we determined the importance score for this feature by averaging the differences of the OOB errors after the overall decision trees were permuted [37]. The importance score produces a measure of the importance of the variables by comparing how much the OOB error of the estimate increases when a variable is permuted, while all other variables are left unchanged. We used this procedure of determining an importance score to select features to determine the relative importance of each variable in the regression.

In this study, we used the training dataset as input data for the RF algorithm for 100 trials to determine the relative importance of all the variables (reflectance of wavelengths) in each trial. Based on the result of each trial, we sorted the variables and separated the variables into varying degrees of importance. According to the cumulative times, we evaluated three sets of wavelengths and defined the top 1st, 5th, and 10th percentile essential wavelengths for subsequent wavelength filtering.

2) 1D-CNN Algorithm: CNNs is a special kind of multilayer neural network for deep learning architectures. A typical CNN usually stacks a series of convolutional layers with nonlinear activation functions and down-sampling pooling layers, accompanied by local connections and shared transform matrices for the artificial neural network. This characteristic makes CNN extract hierarchical features better than traditional methods and makes learning their features automatically instead of coming from a manual design. The one-dimensional CNN (1D-CNN) is a specific form of CNN that adopts a 1-D vector (such as a time-series and spectrum dataset) as input data. 1D-CNN has been successfully used to detect machine damage, recognize human activity, and classify electroencephalography [45]–[47]. In our study, a "LeNet-5"-like 1D-CNN was composed of two convolutional layers, two pooling layers, one flattened layer, and three fully connected layers. Fig. 3 illustrates the architecture of our 1D-CNN algorithm. The roles and parameter selection of each layer in our study were as follows.

1) The input data for the algorithm were the hyperspectral data without preprocessing.
2) The convolutional layers extracted features from the input feature map using the convolution kernels with different sizes and amounts. Each convolution kernel detected specific features on the input feature map to fulfill the weight-sharing on the same input feature map. We supposed the matrix x_j^l $(l-1)$ was an input of the l layer so that the formula for the 1D-CNN algorithm was

$$x_j^l = f\left(b_j^l + \sum_i^M k_{ij}^l * x_i^{l-1}\right) \tag{3}$$

where x_j^l was the jth feature map of the current lth layer; k_{ij}^l represented the convolution kernels; j denotes the number of kernels; M represented the number of input feature maps; b was the bias corresponding to the kernel; $f(\cdot)$ was the activation function; and $*$ denotes the vector convolution.
3) The pooling layer reduced the dimensionality of the feature map while retaining primary information from the input data. It reduced the computational cost by reducing the number of parameters and prevented overfitting;

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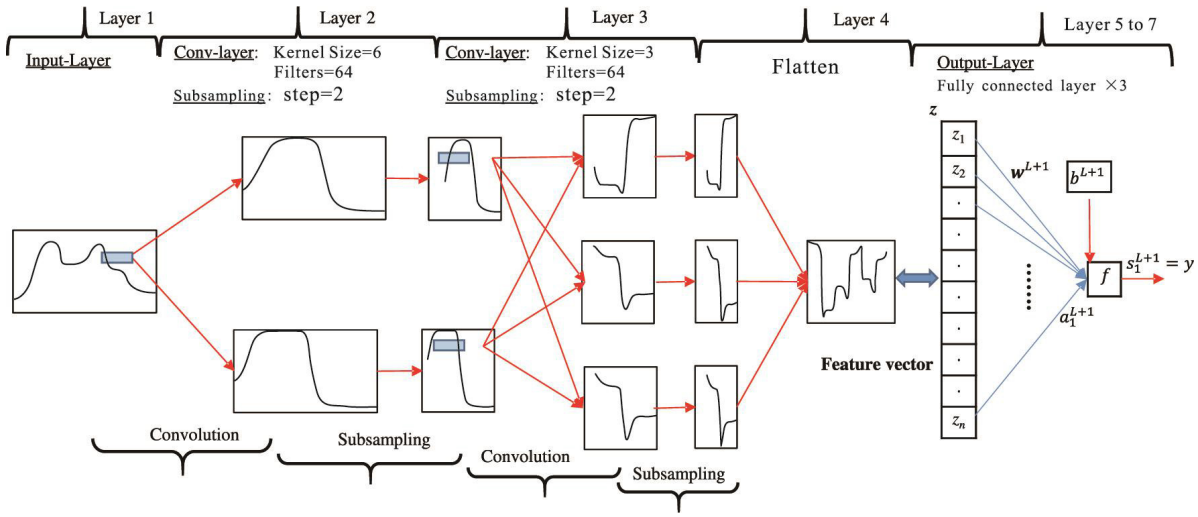


Fig. 3. Architecture of the optimized 1D-CNN retrieval model.

TABLE II
OPTIMAL PARAMETERS FOUND FOR THE 1D-CNN MODEL ALGORITHM

Layer type	Kernel size	Filters	Activation
Convolutional	6	64	ReLU
Maxpooling	2	-	-
Convolutional	3	64	ReLU
Maxpooling	2	-	-
Flatten	-	-	-
Fully-connected	-	32	ReLU
Fully-connected	-	32	ReLU
Fully-connected	-	1	Linear

therefore, the overall performance and accuracy of the network increased. The type of pooling that we used was max-pooling, where we selected the maximum value from the pooled field.

4) The flattened layer was a layer used to convert the output of the upper layer into a 1-D vector that was fed into the fully connected layer.
5) The fully connected layer contained numerous neurons connected to all nodes in the preceding layers. In this study, the third fully connected layer had one neuron, which acted as the output layer.
6) The third fully connected layer used linear activation, and all the other layers used ReLU activation functions.

To fully utilize the 1D-CNN algorithm to automatically extract features from the hyperspectral data, we used a wide convolution kernel (with its size equal to 6) in the first convolution layer to obtain the frequency domain information in the hyperspectral data. We used a smaller convolution kernel (with its size equal to 3) in the subsequent layers to obtain more detailed features. In Table II, we present the other optimal parameters we found for the 1D-CNN model.

We used the statistical platform R (v4.0.3; The R Foundation for Statistical Computing) for the data pretreatment and the RF and 1D-CNN model, the "prospectr," "randomForest," and "keras" packages on an x86-64 PC with AMD Ryzen 7 4800H CPU, 16-G RAM, and NVIDIA RTX 2060 GPU in a Windows 10 environment.

During network training, we initialized parameters such as weight and bias. We first initialized the network by Glorot Initialization for the kernel. Then, we used zero initialization for the bias. We incorporated the Adam optimizer, a mini-batch stochastic gradient descent algorithm, for training with an initial learning rate of 0.001. The training was complete when the admissible conditions for errors were met for the weights and bias updating. We used $\alpha = 0$ in the ReLU function. These weights and bias were then updated through the training processes. We trained the network with a batch size of 16 with a maximum epoch of 200.

E. Evaluating the Dataset and Accuracy for the Retrieval Model

We applied a sample classification strategy to ensure the consistency of the water quality of samples in different datasets: we sorted all samples according to their WQI value, and then divided the first 380 samples according to a ratio of 4:1 in order into the training dataset and the validation dataset. We divided the last two samples (No. 381 and No. 382) into training and validation datasets, respectively. Finally, we separated 306 training samples and 76 validation samples.

In order to reduce the interference caused by the randomness of a single model and obtain the stable performance of the RF and 1D-CNN models, we trained all models that we adopted on their respective training subsets, and we evaluated the models on the respective validation subsets under 100 trials. We averaged the predicted values obtained from the 100 trials and defined their averages to be the final prediction result of the models.

For the validation, we evaluated the accuracy of the models with the coefficient of determination (R^2), the root-mean-square error of validation (RMSE), the mean absolute percentage error (MAPE), and the ratio of performance to the

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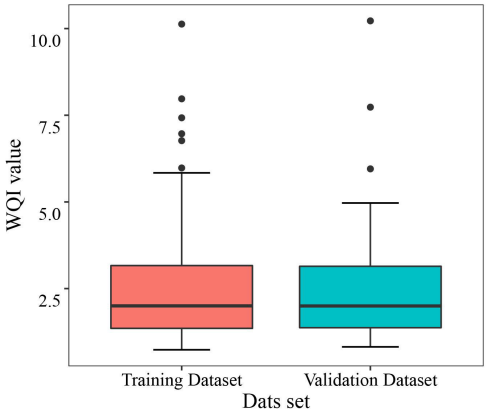


Fig. 4. Characteristics of training and validation.

interquartile range (RPIQ) [48]. These statistics are defined as

$$R^2 = 1 - \frac{\sum_{i=1}^{n} \left(w_i - w_i'\right)^2}{\sum_{i=1}^{n} \left(w_i - \bar{w}_i\right)^2} \tag{4}$$

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{n} \left(w_i - w_i'\right)^2}{n}} \tag{5}$$

$$\text{RPIQ} = \frac{Q_3 - Q_1}{\text{RMSE}} \tag{6}$$

where n is the number of samples; w_i is the measured WQI value of the sample i; w_i' is the predicted WQI value of the sample i; and $\bar{w}_i$ is the average of the measured WQI values. Q_3 and Q_1 are the third and first quartiles of the measured WQI values. In general, having a large R^2 and small RMSE means that the prediction is good. Furthermore, having larger RPIQ values means that the model predicts well.

III. Results and Discussion

A. Water Quality and Spectrum Features

1) Water Quality Features of Samples: We found that the water quality of the collected water samples varied greatly. The measured COD ranged from 5 to 72 mg·L^{-1} with a mean of 13 mg·L^{-1}. TP ranged from 0.02 to 0.95 mg·L^{-1} with a mean of 0.16 mg·L^{-1}, and the NH_4^+-N concentration ranged from 0.027 to 5.84 mg·L^{-1} with a mean of 0.90 mg·L^{-1}. The SS concentration varied from 5 to 351 mg·L^{-1}, and its mean was 39 mg·$^{-1}$. Based on the four parameters' monitoring results, we used (1) to calculate the WQI of each sample.

The WQI values of the training dataset and validation dataset varied from 0.734 to 10.223, with a mean 2.450, and the first Qu., median, and third Qu. numbers were 1.352, 2.001, and 3.183, respectively. The WQI values and the boxplots for the training and validation set (Fig. 4) show that the water quality characteristics of the training and validation datasets were similar, indicating that our sample classification strategy and the results were reasonable.

In order to further reveal the impact of different water quality parameters on the WQI value, we counted the difference in the contribution rate of four parameters to the WQI value of each sample. Table III shows that in different samples, the contribution rate of each parameter to WQI varied greatly. On the whole, no water quality parameter had an absolute dominant influence on the WQI. For all samples, the average contribution rates of TP and SS to WQI were relatively high (33.0% and 27.5%, respectively). Nevertheless, the average contribution rates of the four parameters to WQI were relatively close, indicating the diversity of water quality. The results also show the rationality of using these four parameters to participate in calculating WQI.

TABLE III
Comparison of the Contribution Rate of Each Water Quality Parameter in WQI Calculation

Water quality parameter	Contribution rate in WQI of each sample		
	Min	Max	Mean
COD	2.1%	79.7%	27.5%
TP	3.9%	47.1%	19.1%
NH_4^+-N	3.4%	65.2%	33.0%
SS	1.1%	57.0%	20.4%

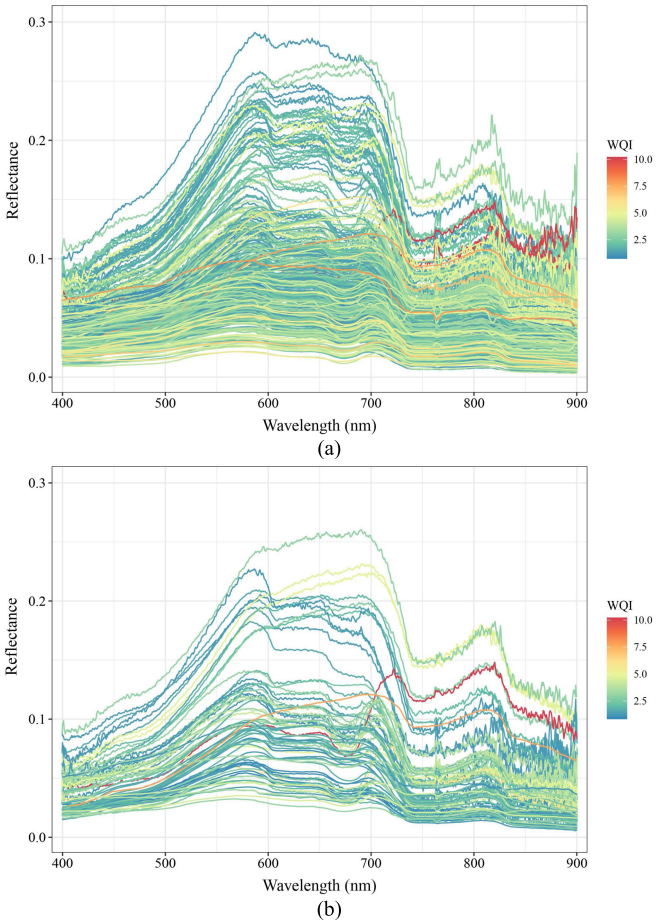


Fig. 5. Hyperspectral features of (a) training dataset and (b) validation dataset.

2) Hyperspectral Feature of Samples: Fig. 5 shows the original spectrum of training and validation samples. The spectrum reflectance of different water samples varied considerably, especially in the wavelength range of ~570 to 700 nm, revealing the spectrum complexity of the water samples. The

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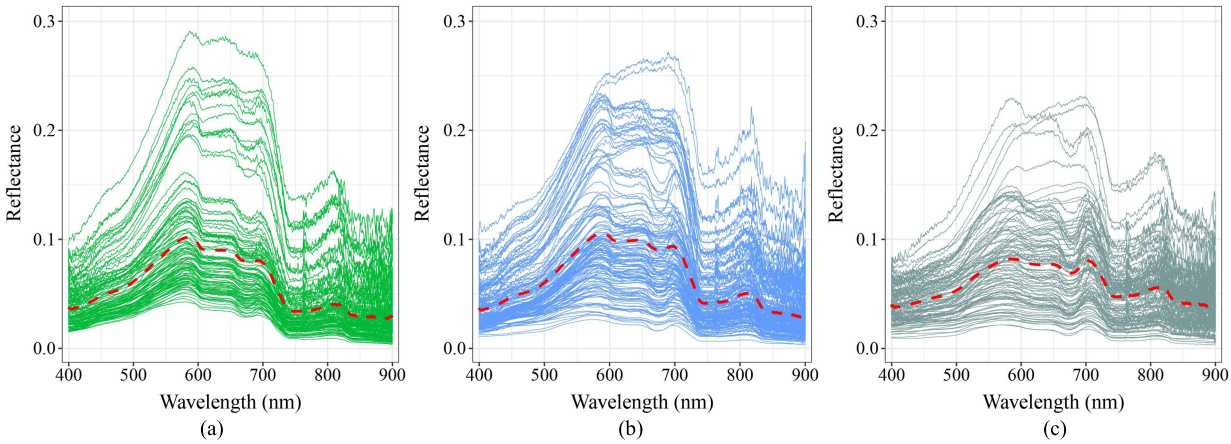


Fig. 6. Spectrum reflectance curves of different categories of WQI (a) samples with WQI < 1.5, (b) samples with 1.5 < WQI < 3, and (c) samples with WQ > 3. The red dotted curves are the mean reflectance of each category.

spectrum reflectance of the monitored rivers was relatively low in the range of 400–500 nm, which would have been mainly due to the absorption of chlorophylls and CDOM. The spectrum reflectance of the water rosed as wavelengths increased, and there were prominent reflection peaks near 585, 710, and 810 nm, and noticeable reflection troughs near 670 and 760 nm.

According to previous related studies, the reflection peaks of surface water near 585 and 710 nm were mainly caused by combined effects of backscattering and weak water absorption. Furthermore, the reflection peaks near 810 nm are due to the small absorption coefficients of water molecules, the scattering effect of suspended particles, and the reflection troughs near 670 and 760 nm caused by the absorption of algae and water molecules, respectively. The samples' hyperspectral features show that the river water's spectrum reflectance would have mainly been affected by suspended particulate matter and chlorophyll-a. Generally, most samples' overall hyperspectral characteristics could be described as a "three peaks and two valleys" morphology. However, this morphology feature was mainly reflected in samples with lower WQI. For the samples with higher WQI, the hyperspectral reflectance curve between 600 and 700 nm was almost straight, with a steady increase instead of an emission peak.

To examine how the spectrum data from different types of rivers varied, we classified the water samples into three categories according to their frequency distributions which represented high water quality (WQI < 1.5), medium water quality (1.5 < WQI < 3), and poor water quality (WQI > 3). We plotted the spectrum data for each category and shown it in Fig. 6. The water reflectance measured at different water quality levels was highly consistent in magnitude and shape, showing no significant differences in the spectrum characteristics of the samples. In the 400–710-nm wavelength regions, the plotted curves in Fig. 6(c) reveal that water bodies with higher WQI had relatively low reflectivity. Conversely, when the water had low WQI, the reflectance spectra of the water in Fig. 6(a) were higher, which means that the spectrum reflectance and water quality were negatively correlated in this band region. However, at wavelengths greater than ∼710 nm, this relationship between WQI and reflectance had the opposite trend: the samples with lower WQI values had low reflectance. These significant changes in water reflectance characteristics for different wavelength intervals below and above ∼700 nm could help identify the essential wavelength for the retrieval model.

B. Retrieval Using Full-Spectrum Data

When we used full-spectrum data to input the retrieval models, the 1D-CNN algorithm performed better than the RF algorithm. 1D-CNN had an R^2 of 0.78, and RF had an R^2 of 0.65 (see Fig. 7). Specifically, the underestimations have been found at the higher end (>3.7) for RF algorithm. In contrast, the 1D-CNN algorithm had good retrieval accuracy for samples with a different range of WQI. In addition, the 1D-CNN algorithm had better retrieval results for samples with extreme high measured WQI (>6), while the RF algorithm had lower retrieval results for these samples. Because samples with high measured values usually have more complex and diverse spectrum characteristics, this result indicates that the 1D-CNN algorithm is more effective and accurate when extracting spectrum features.

C. Sensitivity Analysis of the Retrieval Models to Data Preprocessing

The spectrum reflectance curves in Fig. 6 reveal that hyperspectral reflectance was noisy at longer (greater than 750 nm) wavelengths. In order to examine whether this spectrum noise had an impact on the retrieval model, we applied for a zero-differentiation order, quadratic polynomial Savitzky–Golay filter (with 9, 19, 29, 39, and 49 bands as filter lengths, respectively) to preprocess the data. Fig. 8 shows the reflectance spectrum before and after Savitzky–Golay processing of one representative sample. It can be seen that the filtering process dramatically reduced the reflectance noise between 750 and 900 nm, and smoothed the reflectance spectrum between 400–480 and 580–700 nm.

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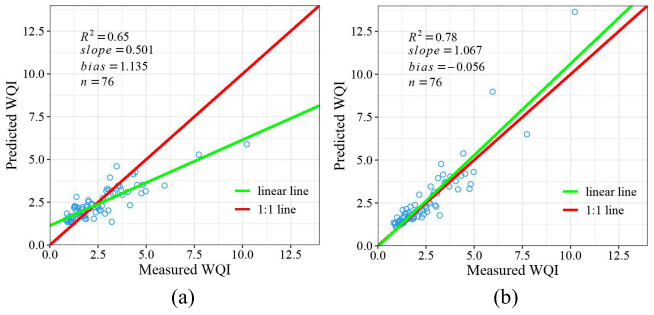


Fig. 7. Observed WQI values vs. the predicted WQI values using full-spectrum data with (a) RF and (b) 1D-CNN models.

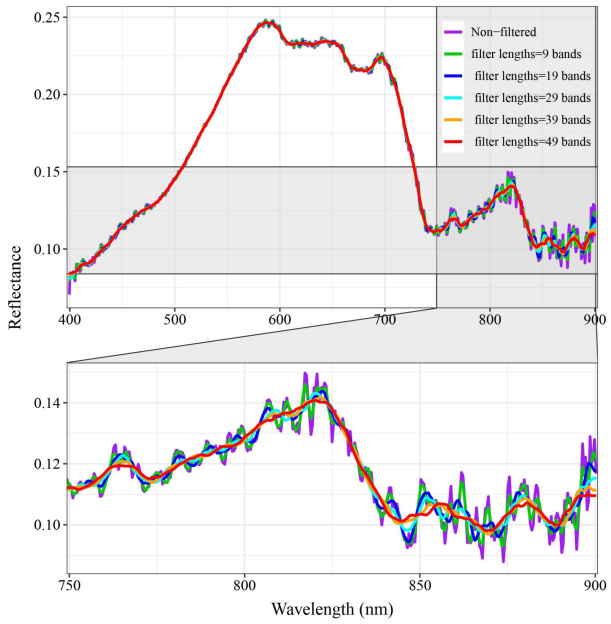


Fig. 8. Spectral reflectance curves of representative sample by smoothing on different wavelengths. The inset shows the 750–800-nm region.

Then, we cross-used the smoothed preprocessed data as training or validation datasets, respectively, into the established RF and 1D-CNN retrieval models. The results are listed in Table IV. A visual comparison show that, whether applied to the training dataset or the validation dataset, the Savitzky–Golay smoothing of the spectrum data had minimal impact on the accuracy of the two models.

On the whole, the retrieval accuracy will slightly decrease or unchanged when the model uses preprocessed data as the training dataset or validation dataset. However, there seems to be no clear rule for the impact of the different filter lengths. Compared with the smoothed data, the unprocessed data were considered to have spectrum noise. Therefore, the above experimental results show that the noise of the original hyperspectral reflectance data has a limited impact on the RF and 1D-CNN algorithms.

D. Determining Essential Wavelengths Using the Random Forest Algorithm

Although the 1D-CNN algorithm achieved better results than the RF algorithm, it was challenging to interpret the results from the neural network. However, determining an essential variable can explain the retrieval mechanism and might help reduce data dimensionality. Fig. 9 demonstrates the importance determined by RF of all wavelengths at different levels. Under different percentile ranges, the selected band ranges were nearly the same. Moreover, the result shows that the wavelengths with the highest mean decrease in accuracy are predominately in the yellow light region (580–590 nm). Additional essential wavelengths are near 722-, 751-, 821-, and 830-nm wavelengths.

TABLE IV
RETRIEVAL RESULTS USING PREPROCESSED VALIDATION DATASETS WITH DIFFERENT FILTER LENGTHS

Data process		RMSE	
Training dataset	Validation dataset	Random Forest	1D-CNN
Non-filtered	Non-filtered	0.939	0.741
Non-filtered	9 bands	0.935	0.768
	19 bands	0.963	0.767
	29 bands	0.971	0.766
	39 bands	0.968	0.765
	49 bands	0.970	0.764
9 bands	Non-filtered	0.997	0.753
19 bands		0.980	0.749
29 bands		0.971	0.796
39 bands		0.961	0.748
49 bands		0.960	0.797

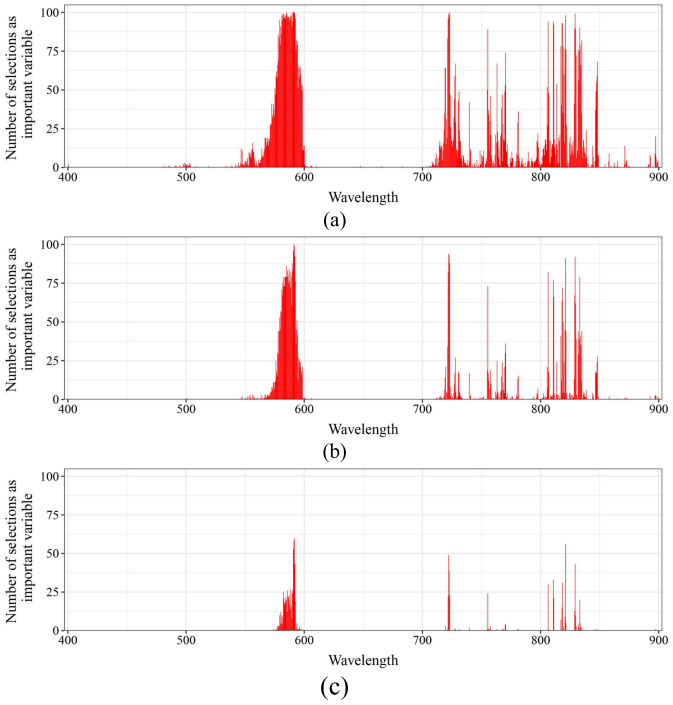


Fig. 9. Varying level of wavelength importance determined by RF. (a) top 10th percentile, (b) top 5th percentile, and (c) top 1st percentile.

E. Retrieval Using Essential Wavelength Data

To explore the different effects of various numbers of essential wavelengths on a model, we used the reflectance data

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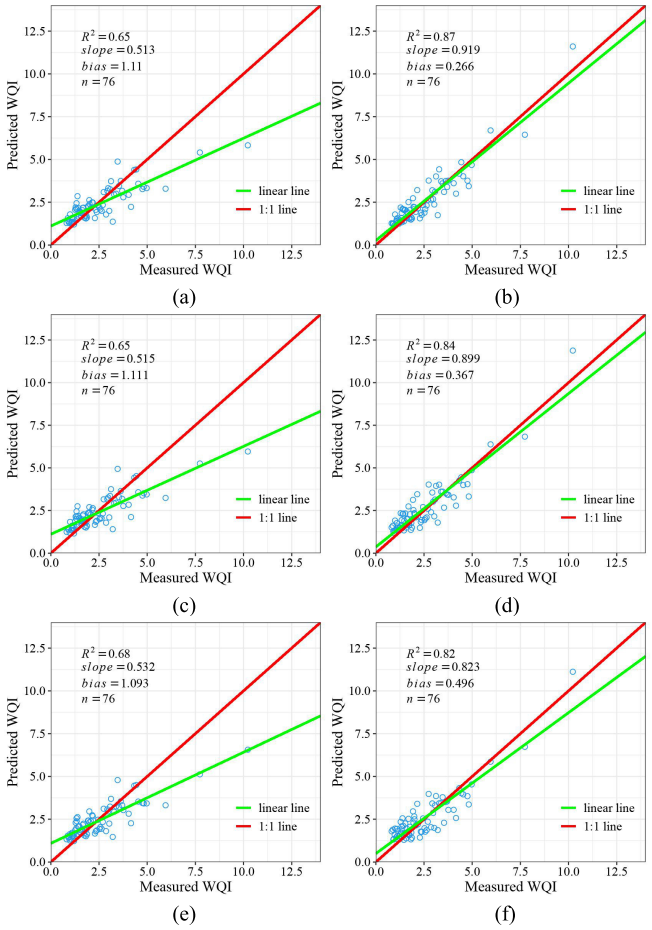


Fig. 10. Observed WQI values vs. the predicted values using the top 10th (a and b), 5th (c, d), and 1st (e and f) percentile essential wavelengths as input data for the RF (a, c, and e) and 1D-CNN (b, d, and f) algorithms.

corresponding to the top 1st (96 bands in total), 5th (280 bands in total), and 10th (484 bands in total) percentile essential variables as input data for the RF and 1D-CNN algorithms.

Fig. 10 illustrates that using various wavelength combinations (after prescreening) as RF and 1D-CNN algorithm input variables achieved better results than using the full-spectrum data as the model input. However, sensitivity to the change of essential wavelength data was different for the two algorithms.

For the RF algorithm, the accuracy of the retrieval results using essential wavelength data was unchanged or even improved as the number of input variables decreased. Using 96 filtered bands, accounting for only 8.4% of the full bands, the RF model was still stable and gained the best accuracy. For the 1D-CNN algorithm, when we used the three essential wavelength combinations as input, the model achieved better results than the full-spectrum data.

Nevertheless, when using the tenth percentile essential wavelength data as input, the 1D-CNN model performed best. The retrieval accuracy decreased slightly as the number of variables decreased, indicating that a certain number of variables was necessary to maintain the effectiveness of the 1D-CNN algorithm. We conclude that reducing excessive dimensionality is not conducive to maintaining the model's effectiveness.

TABLE V
ACCURACY RESULTS OF THE FOUR DIFFERENT VERSIONS OF THE RETRIEVAL MODELS

Data combinations	Random Forest			1D-CNN		
	RMSE	RPIQ	Training time (s)	RMSE	RPIQ	Training time (s)
full-spectrum	0.939	1.885	5.62±0.17	0.741	2.389	75.71±2.96
top 10th percentile	0.943	1.877	3.10±0.07	0.574	3.082	33.75±2.63
top 5th percentile	0.942	1.880	2.62±0.13	0.646	2.739	23.28±0.84
top 1st percentile	0.905	1.955	1.67±0.07	0.681	2.601	14.08±0.84

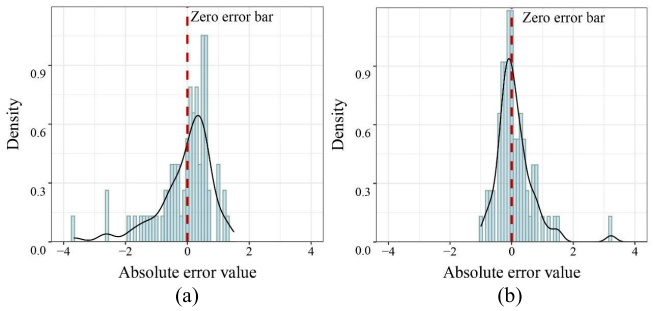


Fig. 11. Distribution of absolute error of the retrieval results of (a) best RF model and (b) best 1D-CNN model.

F. Model Comparison and Discussion

Table V presents more details of the accuracy results for the models using different wavelength combinations as input. The results suggested that regardless of the type of data we used as input, the 1D-CNN model performed better across all properties and produced more robust predictions. The best 1D-CNN model (using the top tenth percentile data as input) with a 0.574 RMSE is 22.5% lower than the full-spectrum data and 36.6% lower than the best RF model (using the top 1st percentile data as input). We plotted the absolute error of the retrieval results of the best RF model and the best 1D-CNN model in Fig. 11. Compared with the best 1D-CNN model, the absolute error distribution of the predicted value of the best RF model was less concentrated.

Further analysis of Figs. 7 and 10 shows that the difference in retrieval capabilities for samples with extremely high WQI was the main reason that affected the accuracy of the RF and 1D-CNN algorithms. The absolute errors of the best RF model for the three samples with extreme high WQI in the validation datasets were 2.63, 2.60, and 3.66, respectively, while the corresponding results of the best 1D-CNN model were 0.74, 1.29, and 1.37, indicating that the 1D-CNN model performed much better on high WQI samples. If removing three samples with extreme high WQI, the accuracy of the best 1D-CNN model (with a 0.536 RMSE) was still much better than the best RF model (with a 0.693 RMSE). The RPIQ of all versions of the 1D-CNN models was more significant than for the RF

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models, and the best RPIQ was 3.082, which was predicted excellently.

Table V also reports the training and test time of the RF and 1D-CNN models with different datasets. The training speed of the models increased significantly as the amount of input data decreased, and the RF algorithm demonstrated superior computational efficiency in model training. Unlike 1D-CNN, which needs to calculate a large number of the deep neural network parameters, the RF algorithm selected a random subset that only included a part of features from all available input features of each internal decision tree to participate in modeling, which significantly reduced the complexity of calculations.

The results reveal that a small number of essential wavelengths played a vital role in both retrieval models. Therefore, selecting essential wavelengths is significant to model improvement. For the RF retrieval model, dimensionality reduction was significant in reducing data redundancy and improving retrieval accuracy. Nevertheless, the 1D-CNN algorithms had a different applicable effect for reducing dimensionality. Because convolution and pooling operations rapidly reduce data dimensionality, it is necessary to maintain a certain number of input variables in the 1D-CNN network to ensure the robustness of the model. After selecting the essential wavelength, the optimal input spectrum wavelengths used for modeling decreased from 1136 to 96 (for RF) and 484 (for 1D-CNN), while the retrieval effect of the models did not decrease significantly. This result demonstrates that determining an essential wavelength can screen for the spectrum information interval, which has the most immediate effect on the water quality of rivers. The essential wavelength selection method can help simplify the model while ensuring the model's prediction accuracy and improving modeling efficiency. This is an encouraging result for application to hyperspectral data, although determining the optimal wavelength combination for the input data still requires much work for the 1D-CNN algorithm.

G. Application Prospects and Challenges

In recent studies, establishing and comparing water quality retrieval models based on *in situ* hyperspectral data and then applying the optimized model to hyperspectral images is a common technical framework. Especially, with the development of UAV hyperspectral measurement technology, many researchers have successfully estimated individual water quality parameters following this framework and formed continuous spatial distribution of water quality parameters. Although we take WQI as the retrieval target instead of conventional water quality parameters in this study, the method and process of establishing the relationship between WQI and water reflection spectrum characteristics are the same. Therefore, it should be feasible to apply this model to images. Results of this study show that the noise of the original hyperspectral reflectance has a limited impact on the RF and 1D-CNN algorithms. However, when the algorithms are applied to the hyperspectral images with more noises (radiometric noises and atmospheric correction noises), is the retrieval accuracy still not affected by the spectrum noise? This will be the key issue that needs to be paid attention to when applying the models to images or other scenarios [52]–[54]. The performance of the RF and 1D-CNN algorithms by adding random noises within ±5% and ±3% of original reflectance is also tested, and the results also show that the addition of random noise has little effect on retrieval accuracy (see Appendix A for details). This suggests that the proposed algorithms have reasonable robustness for further applications. Although we take urban rivers as the research objects in this study, the reflectance values and spectrum characteristics of samples are similar to other researches on inland water [10], [49], large rivers [3], reservoirs [50], [51], and coastal waters [52], [53]. Therefore, it is reasonable to believe that the algorithm obtained in this study is not limited to its application in urban rivers. We will try to apply the method proposed in this study to remote sensing images and other water types in future work.

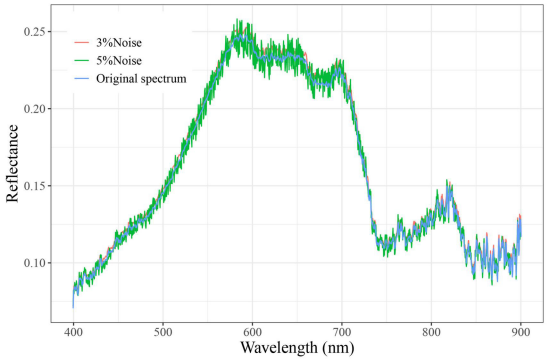


Fig. 12. Sample of noise-added spectral reflectance: original one, original one ±3% noise, and original one ±5% noise.

In this experiment, we have reported the calculation cost of the two algorithms, but had not tested the impact of the GPU computing acceleration power on improving the training speed of the 1D-CNN algorithm. The use of more specific computing acceleration for the 1D-CNN algorithm would be one of our future works.

IV. Conclusion

This article reveals that *in situ* hyperspectral data can identify spectrum characteristics that are optimal for determining water quality and can provide a practical step for water quality mapping. The hyperspectral features of samples show that the spectrum reflectance of river water is mainly affected by suspended particulate matter and chlorophyll-a, and there is a specific correlation between the spectrum reflectance of different bands and the WQI. This feature of water samples makes it possible to use spectrum reflectance to distinguish water quality levels with different WQI values.

Based on RF and 1D-CNN algorithms, the overall accuracy demonstrated that both machine learning algorithms performed well for WQI retrieval using hyperspectral reflection data. Furthermore, both models are robust and insensitive to input data smoothing. The 1D-CNN model can identify the reflectance spectrum of rivers that have poor water quality. Meanwhile, the 1D-CNN algorithm generally performed better than the

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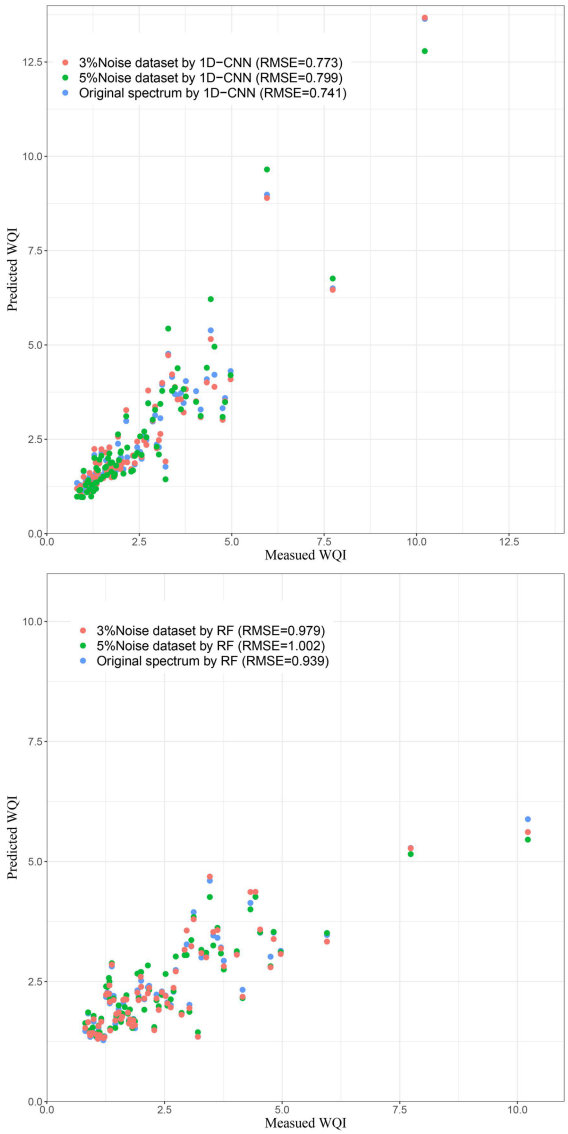


Fig. 13. The observed WQI values versus the predicted WQI values by different datasets using 1D-CNN and RF algorithm.

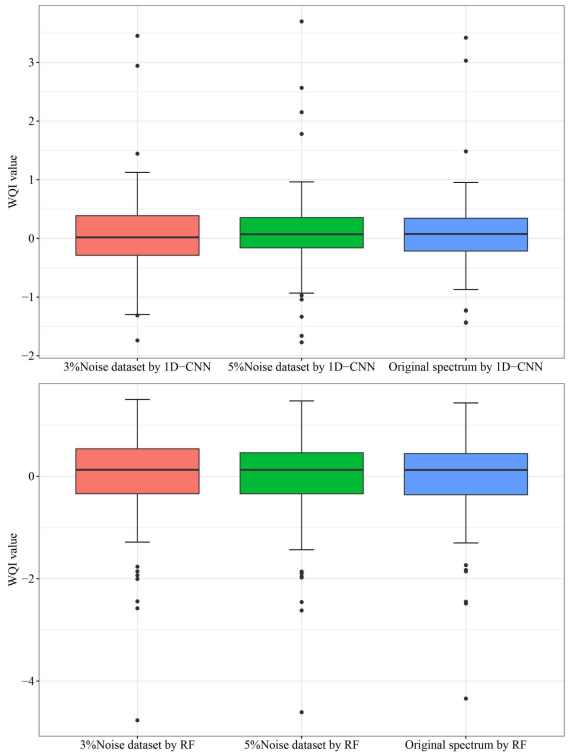


Fig. 14. Boxplots of absolute error of the retrieval results for validation samples by different datasets using 1D-CNN and RF algorithm.

RF algorithm when full-spectrum data or essential wavelength data were used. Furthermore, the 1D-CNN algorithm we used in this study is relatively basic and does not need a complicated optimization process. Our results showed that the simple 1D-CNN mode performed best, having an RMSE of 0.574 and R^2 of 0.87. These results reveal that hyperspectral data combined with the 1D-CNN algorithm can be used for rapidly and accurately identifying water quality of urban rivers without complex preprocessing of input data.

The essential variable assessment by the RF algorithm is a necessary step for improving retrieval accuracy. The methods we used in this study can successfully reduce the number of wavelengths to a significant few to achieve sufficient accuracy.

Overall, this study illustrates that the 1D-CNN algorithm is robust and accurate for predicting the WQI of urban rivers using hyperspectral data. While 1D-CNN is an excellent algorithm, future work might focus on parameter and architecture optimization for higher accuracy. If *in situ* measurements and the 1D-CNN algorithm complement airborne and satellite data, then the enhanced spectrum and spatial data resolution might provide a better platform for comprehensive water quality assessment.

APPENDIX A

To test the robustness of the machine learning algorithms, random noises within $\pm 5\%$ and $\pm 3\%$ of corresponding bands are added to the original spectrum (see Fig. 12). Then, the results derived through the machine learning models are analyzed. The results are shown in Figs. 13 and 14.

The scatter plots of the predicted values of validation samples obtained with different noise-added datasets also show that the addition of random noise has little effect on the retrieval result on each sample (Fig. 13).

We analyzed the absolute error between the predicted value and the measured value of the validation samples when using different noise-added datasets. It can be seen from Fig. 14 that there is no significant difference in the retrieval errors when using different noise-added datasets. In summary, we conclude that the addition of random noise has little effect on retrieval accuracy, indicating that the two machine learning algorithms are robust to random noise.

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